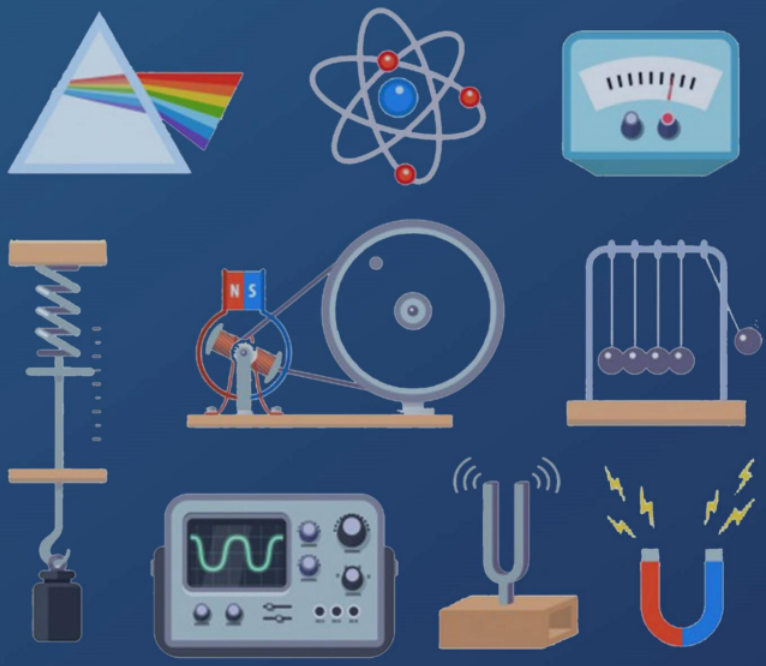
PHYSICS

Alternating Current



SHEET

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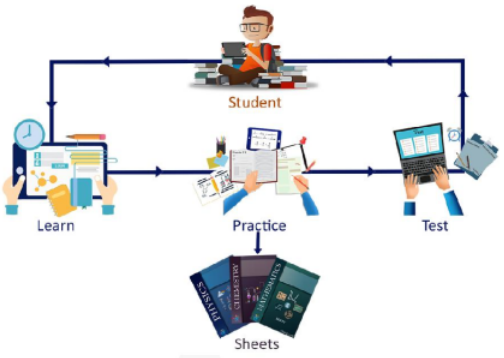
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex 1	Ex 1A	Ex 2	Ex 2A
1	Alternating Current-Introduction				
2	Average Value and RMS Value	1	1	1	1,2
3	AC Voltage Applied to a Resistor	2			
4	AC Voltage Applied to a Capacitor	3	10	2	3
5	AC Voltage Applied to an Inductor	4	2		
6	Phasors and Phasor Diagram	5	3		
7	Impedance		11	3	4,5
8	AC Voltage Applied to Series LR circuit	6	4,12	4,5	6
9	AC Voltage Applied to Series RC circuit		9	6	
10	AC Voltage Applied to Series LCR circuit	7	5	7	7,10-12
11	Power in AC circuits-The Power Factor	8,9	13	8,9	8
12	Resonance Condition	10,11	14,15	10,11	
13	LC Oscillations	12		12	14
14	Transformers	13,14	6	13	15
15	After complete chapter	15,16	7,8	14	9,13

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Average Value and RMS Value

Q1. The r.m.s. value of alternating current is 10 ampere having frequency of 50 Hz. The time taken by the current to increase from zero to maximum and the maximum value of current will be :-

(A) 2×10^{-2} s and 14.14 A

(B) 1×10^{-2} s and 7.07 A

(C) 5×10^{-3} s and 7.07 A

(D) 5×10^{-3} s and 14.14 A

AC Voltage Applied to a Resistor

Q2. In a resistance of 25 Ω A.C. is passed to produce heat at the rate of 250 watt. The value of current in the resistance will be :–

(A) 0.316 A (B) 1 A

(C) 3.16 A (D) 10 A

AC Voltage Applied to a Capacitor

Q3. Let f = 50 Hz, and C = 100 μF in an AC circuit containing a capacitor only. If the peak value of current in the circuit is 1.57 A at t = 0, then equation for AC Voltage will be :

(A) E = 50 sin (100πt - π/2)

(B) E = 100 sin (50πt)

(C) E = 50 sin 100πt

(D) E = 50 sin (100πt + π/2)

AC Voltage Applied to an Inductor

Q4. An a.c. source of voltage V and of frequency 50Hz is connected to an inductor of 2H and negligible resistance. A current of r.m.s. value I flows in the coil. When the frequency of the voltage is changed to 400 Hz keeping the magnitude of V same, the current is now

(A) 8I in phase with V

(B) 4I and leading by 90 from V

(C) I/4 and lagging by 90^0 from V

(D) I/8 and lagging by 90^0 from V

Phasors and Phasor Diagram

Q5. A capacitor of capacitance 100 μF and a resistance of 100Ω is connected in series with AC supply of 220V, 50Hz. The current leads the voltage by :-

(A) $\tan^{-1}\left(\frac{1}{2\pi}\right)$ (B) $\tan^{-1}\left(\frac{1}{\pi}\right)$

(C) $\tan^{-1}\left(\frac{2}{\pi}\right)$ (D) $\tan^{-1}\left(\frac{4}{\pi}\right)$

AC Voltage Applied to Series LR circuit

Q6. In an A.C. circuit resistance and inductance are connected in series. The potential and current in inductance is :-

(A) $V_0 \sin \omega t, \frac{V_0}{L\omega} \sin \omega t$

(B) $V_0 \sin \omega t, \frac{V_0}{L\omega} \sin\left[\omega t + \frac{\pi}{2}\right]$

(C) $V_0 \sin\left[\omega t + \frac{\pi}{2}\right], \frac{V_0}{L\omega} \sin \omega t$

(D) $V_0 \sin\left[\omega t + \frac{\pi}{2}\right], \frac{V_0}{L\omega} \sin\left[\omega t - \frac{\pi}{2}\right]$

AC Voltage Applied to Series LCR circuit

Q7. A circuit contains R, L and C connected in series with an A. C. source. The values of the reactances for inductor and capacitor are 200Ω and 600Ω respectively and the impedance of the circuit is Z_1. What happens to the impedance of the same circuit if the values of the reactances are interchanged :-

(A) The impedance will remain unchanged

(B) The impedance will increase

(C) The impedance will decrease

(D) Information insufficient

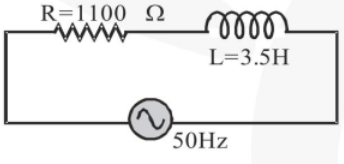
Power In AC Circuits-The Power Factor

Q8. An AC circuit draws 5A at 160 V and the power consumption is 600 W. Then the power factor is :-

(A) 1 (B) 0.75

(C) 0.50 (D) Zero

Q9. For given circuit the power factor is :-



(A) 0 (B) $\frac{1}{2}$

(C) $\frac{1}{\sqrt{2}}$ (D) none of these

Resonance Condition

Q10. An inductance of $1mH$, a condenser of $10\mu F$ and a resistance of 50Ω are connected in series. The reactance of inductor and condensers are same. The reactance of either of them will be :-

(A) 100 Ω (B) 30 Ω

(C) 3.2 Ω (D) 10 Ω

Q11. In LCR circuit, capacitor C is changed to 4C, then what should be the value of L to keep resonance frequency same :-

(A) 2L (B) $\frac{L}{2}$ (C) $\frac{L}{4}$ (D) 4L

LC Oscillations

Q12. Which statement is correct from following-

(a) Inductor store energy in the form of magnetic field

(b) Capacitor store energy in the form of electric field

(c) Inductor store energy in the form of electric and magnetic field both

(d) Capacitor store energy in the form of electric and magnetic field both

(A) a, b (B) a, c

(C) b, d (D) b, c

Transformers

Q13. The number of turns in the primary and secondary of transformer are N_P and N_S respectively. The resistance of the secondary circuit is R_S. Then equivalant resistance of primary circuit :–

(A) $R_s\frac{N_S}{N_P}$ (B) $R_s\left[\frac{N_S}{N_P}\right]^2$

(C) $R_s\frac{N_P}{N_S}$ (D) $R_s\left[\frac{N_P}{N_S}\right]^2$

Q14. The ratio of the secondary to the primary turns in a transformer is 3 : 2 and the output power is P. Neglecting all power losses, the input power must be :-

(A) $\frac{P}{2}$ (B) P

(C) $\frac{2P}{3}$ (D) $\frac{3P}{2}$

After Complete Chapter

Q15. The dimensions of combination $\frac{L}{CVR}$ are same as dimensions of :-

(A) Change (B) Current

(C) Charge^{-1} (D) Current^{-1}

Q16. L, C and R respectively indicate inductance, capacitance and resistance. Select the combination, which does not have dimensions of frequency :-

(A) 1/RC (B) R/L

(C) $1/\sqrt{LC}$ (D) C/L

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. A generator produces a voltage V=240 sin120t volt where V is in volts and t in second. The frequency and r.m.s. voltage are nearly :-

(A) 60 Hz and 240 V

(B) 19 Hz and 120 V

(C) 19 Hz and 170 V

(D) 754 Hz and 170 V

Q2. In a circuit the frequency is $f = \frac{1000}{2\pi}$ Hz and the inductance is 2 henry, then the reactance will be :-

(A) 200 Ω (B) 200 μΩ

(C) 2000 Ω (D) 2000 μΩ

Q3. 200 Ω resistance and 1H inductance are connected in series with an A.C. circuit. The frequency of the source is $\frac{200}{2\pi}$ Hz. Then phase difference in between V and I will be :-

(A) 30° (B) 60°

(C) 45° (D) 90°

Q4. An LR circuit with a battery is connected at t = 0. Which of the following quantities is not zero just after the circuit.

(A) current in the circuit

(B) magnetic field energy in the inductor

(C) power delivered by the battery

(D) emf induced in the inductor

Q5. For a LCR series circuit with an A.C. source of angular frequency ω :

(A) circuit will be capacitive if $\omega > \frac{1}{\sqrt{LC}}$

(B) circuit will be inductive if $\omega > \frac{1}{\sqrt{LC}}$

(C) power factor of circuit will by unity if capacitive reactance equals inductive reactance

(D) current will be leading if $\omega > \frac{1}{\sqrt{LC}}$

Q6. The core of any transformer is laminated so as to -

(A) Make it light weight

(B) Make it robust and strong

(C) Increase the secondary voltage

(D) Reduce the energy loss due to eddy current

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 7 & 8

In an L - C - R series circuit connected to an AC source, $V = V_0 \sin 5\left(100\pi t + \frac{\pi}{6}\right)$.

Given V_R = 40 V, V_L = 40 V and V_C = 10 V. Resistance R = 4Ω.

Q7. Peak value of current in the circuit is :-

(A) $10\sqrt{2}$ A (B) $15\sqrt{2}$ A

(C) $20\sqrt{2}$ A (D) $25\sqrt{2}$ A

Q8. Choose the correct option :-

(A) $L = \frac{1}{25\pi}$ H

(B) $C = \frac{1}{50\pi}$ F

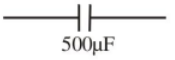
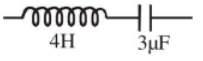
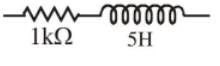
(C) both (A) and (B) are correct

(D) both (A) and (B) are wrong

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q9. Match the following items –

Column-I **Circuit component across an ac source (ω = 200 rad/sec)**	**Column-II** **Phase difference between current and source voltage**
(A) 	(P) $\frac{\pi}{2}$
(B) 	(Q) $\frac{\pi}{6}$
(C) 	(R) $\frac{\pi}{4}$
(D) 	(S) $\frac{\pi}{3}$
(E) 	(T) None of the above

SUBJECTIVE TYPE

Q10. A 15.0 μF capacitor is connected to a 220 V, 50 Hz source. Find the capacitive reactance and the current (rms and peak) in the circuit. If the frequency is doubled, what happens to the capacitive rectance and the current ?

Q11. A bulb is connected to a mains line of 200 V, 50 Hz. If the bulb draws the same current of 10A from a dc source of 100 V, then find the inductance of the filament of the bulb.

Q12. A coil of reactance 100 Ω and resistance 100 Ω is connected to a 240 V, 50 Hz a.c. supply.

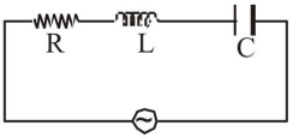
(a) What is the maximum current in the coil?

(b) What is the time lag between the voltage maximum and the current maximum ?

Q13. In given circuit R = 100Ω. If voltage leads current by 60° then find -

(a) Maximum Current supply by source.

(b) Average power

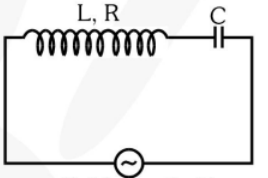


$V = 200\sin 100\pi t$ volt

Q14. A coil, a capacitor and an a.c. source of rms voltage 24V are connected in series, by varying the frequency of the source, a maximum rms current 6A is observed, if this coil is connected to a battery of emf 12V and internal resistance 4Ω, then calculate the current through the coil.

Q15. A coil of resistance R and inductance L is connected in series with a capacitor C and complete combination is connected to a a.c. voltage, Circuit resonates when angular frequency of supply is $\omega = \omega_0$.

Find out relation betwen ω_0, L and C.



Exercise - 2 — *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The current in a discharging LR circuit is given by $i = i_0 e^{-t/\tau}$ where τ is the time constant of the circuit. Calculate the rms current for the period $t = 0$ to $t = \tau$.

(A) $\frac{i_0}{e}\sqrt{\frac{(e^2-1)}{2}}$ (B) $\frac{i_0}{e}\sqrt{\frac{(e^{-2}+1)}{2}}$

(C) $\frac{i_0}{e}\sqrt{\frac{(e+2)}{3}}$ (D) $\frac{i_0}{e}\sqrt{\frac{(e^2-2)}{2}}$

Q2. In an A.C. circuit capacitance of 5μF has a reactance 1000 Ω. The frequency of A.C. will be :–

(A) $\frac{1000}{\pi}$ cycle/s (B) $\frac{100}{\pi}$ cycle/s

(C) 200 cycle/s (D) 5000 cycle/s

Q3. When 100V D.C. is applied across a coil a current of 1A flows through it. When 100V A.C. of 50 Hz is applied to the same coil only 0.5A flows. The inductance of the coil is :–

(A) 0.55 H

(B) 55 mH

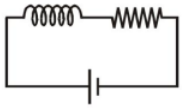
(C) 0.55 mH

(D) 5.5 mH

Q4. A 12 ohm resistor and a 0.21 henry inductor are connected in series to an AC source operating at 20 volts, 50 cycle/second. The phase angle between the current and the source voltage is:

(A) 30° (B) 40° (C) 80° (D) 90°

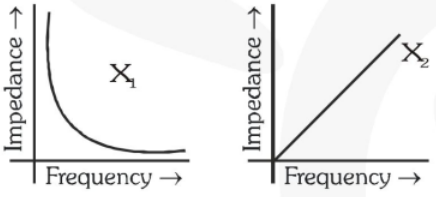
Q5. Time constant of the given circuit is τ. If the battery is replaced by an AC source having voltage $V = V_0 \cos \omega t$, power factor of the circuit will be :-



(A) $\omega\tau$ (B) $\frac{1}{\sqrt{1+(\omega\tau)^2}}$

(C) $\sqrt{1+(\omega\tau)^2}$ (D) None

Q6. The graphs given below depict the dependence of two reactive impedances X_1 and X_2 on the frequency of the alternating e.m.f. applied individually to them. We can then say that :



(A) X_1 is an inductor and X_2 is a capacitor
(B) X_1 is a resistor and X_2 is a capacitor
(C) X_1 is a capacitor and X_2 is an inductor
(D) X_1 is an inductor and X_2 is a resistor

Q7. Which of the following statements is correct, for an LCR series combination having the resonating condition as :-

(A) the current is minimum.

(B) the phase difference between the current and e.m.f. is $\frac{\pi}{2}$.

(C) the impedance is equal to R.

(D) the value of power factor is minimum.

Q8. Power factor of an L–R series circuit is 0.6 and that of a C–R series circuit is 0.5. If the element (L,C, and R) of the two circuits are joined in series the power factor of this circuit is found to be 1. The ratio of the resistance in the L–R circuit to the resistance in the C–R circuit is -

(A) $\frac{6}{5}$ (B) $\frac{5}{6}$ (C) $\frac{4}{3\sqrt{3}}$ (D) $\frac{3\sqrt{3}}{4}$

Q9. A series R - L - C ($R = 10\ \Omega$, $X_L = 20\ \Omega$, $X_C = 20\ \Omega$) circuit is supplied by $V = 10 \sin \omega t$ volt then power dissipation in circuit is :-

(A) Zero (B) 10 watt
(C) 5 watt (D) 2.5 watt

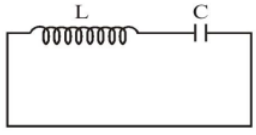
Q10. The self inductance of the motor of an electric fan is 10 H. In order to impart maximum power at 50Hz. It should be connected to a capacitance of :

(A) 2×10^{-6} F (B) 3×10^{-6} F
(C) 10^{-4} F (D) 10^{-6} F

Q11. In a series resonant R-L-C circuit, if L is increased by 25% and C is decreased by 20%, then the resonant frequency will :

(A) Increases by 10%
(B) Decreases by 10%
(C) Remain unchanged
(D) Increases by 2.5%

Q12. In an LC circuit, the capacitor has maximum charge q_0. The value of $\left|\frac{di}{dt}\right|_{max}$ is -



(A) $\frac{q_0}{LC}$ (B) $\frac{q_0}{\sqrt{LC}}$

(C) $\frac{q_0}{LC} - 1$ (D) None of these

Q13. An ideal efficient transformer has a primary power input of 10kW. The secondary current when the transformer is on load is 25A. If the primary : secondary turns ratio is 8 : 1, then the potential difference applied to the primary coil is

(A) $\frac{10^4 \times 8^2}{25}$ V (B) $\frac{10^4 \times 8}{25}$ V

(C) $\frac{10^4}{25 \times 8}$ V (D) $\frac{10^4}{25 \times 8^2}$ V

Q14. An AC voltage source of variable angular frequency ω and fixed amplitude V connected in series with a capacitance C and an electric bulb of resistance R (inductance zero). When ω is increased :

(A) the bulb glows dimmer

(B) the bulb glows brighter

(C) total impedence of the circuit is unchanged

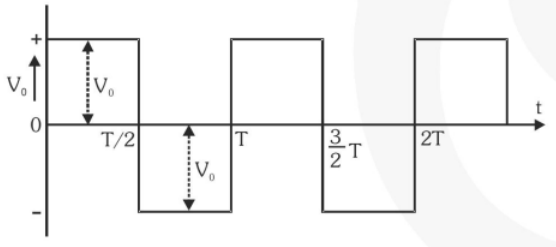
(D) total impedence of the circuit increases

Exercise - 2A — *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The average and rms value of voltage for square wave shown in fig. having peak value V_0 -



(A) $\frac{V_0}{\sqrt{2}}$, $\sqrt{2}V_0$ (B) $\sqrt{2}V_0$, $\frac{V_0}{\sqrt{2}}$

(C) V_0, V_0 (D) Zero, Zero

Q2. A current in circuit is given by $i = 3 + 4 \sin \omega t$. Then the effective value of current is :

(A) 5 (B) $\sqrt{7}$

(C) $\sqrt{17}$ (D) $\sqrt{10}$

Q3. A capacitor of capacity C is connected in A.C. circuit. The applied emf is $V=V_0 \sin\omega t$, then the current is -

(A) $I = \frac{V_0}{C\omega} \sin\omega t$

(B) $I = \frac{V_0}{C\omega} \sin\left[\omega t + \frac{\pi}{2}\right]$

(C) $I = V_0 C\omega \sin\omega t$

(D) $I = V_0 C\omega \sin\left[\omega t + \frac{\pi}{2}\right]$

Q4. A student connects a long air cored - coil of manganin wire to a 100 V D.C. supply and records a current of 25 amp. When the same coil is connected across 100 V, 50 Hz AC. the current reduces to 20 A , the reactance of the coil is :-

(A) 4 Ω (B) 3 Ω

(C) 5 Ω (D) None

Q5. A capacitor has a resistance of 12 ohm and a capacitance of 22 microfarad. It is connected to an AC supply of 80 Hz, then the e.m.f. of the AC supply required to drive a current of 10 virtual ampere is -

(A) $904\sqrt{2}$ volt (B) 913 volt

(C) $\frac{904}{\sqrt{2}}$ volt (D) 452 volt

Q6. For L-R circuit, the time constant is equal to -

(A) twice the ratio of the energy stored in the magnetic field to the rate of the dissipation of energy in the resistance.

(B) the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance

(C) half of the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance

(D) square of the ratio of the energy stored in the magnetic field to the rate of dissipation energy in the resistance

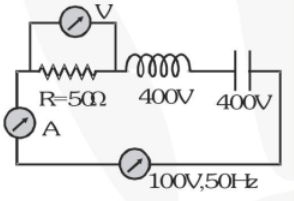
Q7. In given LCR circuit, the voltage across the terminals of a resistance and current will be:-

(A) 400V, 2A

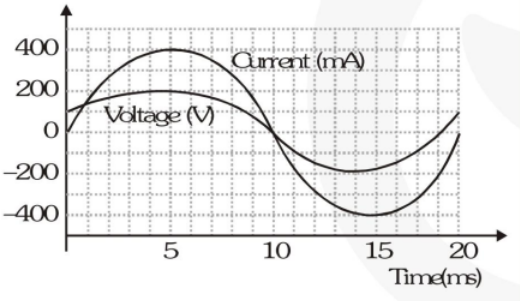
(B) 800V, 2A

(C) 100V, 2A

(D) 100V, 4A



Q8. The given graph shows variation with time in the source voltage and steady state current drawn by a series RLC circuit :-



Which of the following statements is/are correct?

(a) Current lags the voltage

(b) Resistance in the circuit is $250\sqrt{3}\Omega$

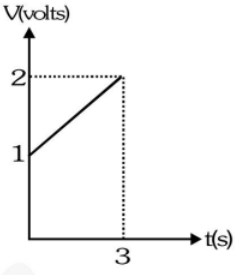
(c) Reactance in the circuit is 250Ω

(d) Average power dissipation in the circuit is $20\sqrt{3}$W

(A) Only a (B) a & b

(C) a, b & c (D) All

Q9. A circuit element is placed in a closed box. At time t = 0, constant current generator supplying a current of 1 A, is connected across the box. Potential difference across the box varies according to graph shown in figure. The element in the box is/are :



(A) resistance of 2Ω

(B) battery of emf 6V

(C) inductance of 2H

(D) capacitance of 0.5 F

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for question nos. 10 to 12

If various elements, i.e., resistance, capcitance and inductance which are in series and having values 1000 Ω, 1μF and 2.0 H respectively. Given emf as, $V = 100\sqrt{2}\sin 1000\,t$ volts.

Q10. Voltage across the resistor is -

(A) 70.7 Volts (B) 100 Volts

(C) 141.4 Volts (D) 270.7 Volts

Q11. Voltage across the inductor is -

(A) 70.7 Volts (B) 100 Volts

(C) 141.4 Volts (D) 270.7 Volts

Q12. Voltage across the capacitor is

(A) 70.7 Volts (B) 100 volts

(C) 141.4 Volts (D) 270.7 Volts

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q13. Instantaneous voltage and instantaneous current in an L-R AC circuit is V = 100 sin (100 t) and $i = 10 \sin\left(100t - \frac{\pi}{4}\right)$. Match the entries of column I with the entries of column II.

Column-I	**Column-II**
(a) R	(p) $(1/10\sqrt{2})$SI unit
(b) X_L	(q) $5\sqrt{2}$ SI unit
(c) L	(r) $10\sqrt{2}$ SI unit
(d) Average power in one cycle	(s) None of these

(A) (a - p), (b - q), (c - r), (d -s)
(B) (a - q), (b - p), (c - s), (d - s)
(C) (a - q), (b - q), (c - p), (d -s)
(D) None of these

SUBJECTIVE TYPE

Q14. Comparing the L–C oscillations with the oscillations of a spring–block system (force constant of spring = k and mass of block = m), the physical quantity mk is similar to _____.

Q15. A transformer having efficiency 90% is working on 100 V and at 2.0 kW power. If the current in the secondary coil is 5A, calculate (i) the current in the primary coil and (ii) voltage across the secondary coil.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
D	C	C	D	B	C	A	B	C	D
11	12	13	14	15	16				
C	A	D	B	D	C				

Ex-1A

1. C **2.** C **3.** C **4.** D **5.** B,C

6. D **7.** A **8.** A

9. (A) → (R) ; (B) → (P) ; (C) → (P) ; (D) → (T) ; (E) → (R)

10. 1/2 , 2 **11.** 0.055H **12.** (a) 2.4 A (b) 2.5 ms.

13. (a) 1 A (b) 50Watt **14.** 1.5 A **15.** $\omega_0 = \frac{1}{\sqrt{LC}}$

Ex-2									
1	2	3	4	5	6	7	8	9	10
A	B	A	C	B	C	C	C	C	D
11	12	13	14						
C	A	B	B						

Ex-2A

1. C **2.** C **3.** D **4.** B **5.** B

6. A **7.** C **8.** D **9.** D **10.** A

11. C **12.** A **13.** C **14.** $\frac{L}{C}$

15. (i) 20 A (ii) 360 V